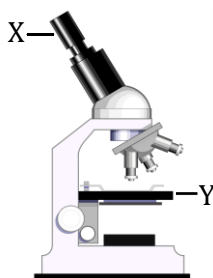




Cell: The Building Block of Life DPP - 01

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

Directions: (Q.1 and Q.2) Observe the given image of compound microscope and answer the question.



- In the given image of compound microscope X is -
 (1) objective lens (2) eye piece
 (3) stage (4) screw
- Object is placed above the -
 (1) X (2) Y (3) Z (4) X and Y both
- Match the Column I with the Column II.

Column - I		Column - II	
A.	Elongated and branched	(p)	Paramecium
B.	Slipper Shaped	(q)	Nerve Cell
C.	Cuboidal	(r)	RBC
D.	Discoidal Shaped	(s)	Germ cells of gonads

Analogy Type Questions

- Germ cells : Cuboidal :: Liver cell : _____.
- Robert Hooke : Cell :: Robert Brown : _____.

Read the following statements and give your answer as true or false.

- The size of an organism is dependent upon the number of cells and not on the size of the cell.
- Muscle cell is spindle shaped.

Read the following statements and fill in the blanks.

- A cell is the _____ and functional unit of life.
- The smallest cell known is that of _____.

Subjective Question

- What is cell theory?

SOLUTIONS DPP - 01

1. Option (2)

In the given image X is eyepiece.

2. Option (2)

Object is placed above the stage (Y).

3. A – (q), B – (p), C – (s), D – (r)

4. Polygonal

5. Nucleus

6. True

7. True

8. Structural

9. PPLO

10. The “cell theory” was formulated by two biologists, M. J. Schleiden (1838), and T. Schwann (1839).

According to them, the cell is the structural and functional unit of all living beings. The cell theory was further expanded by Virchow.

Cell theory, states that

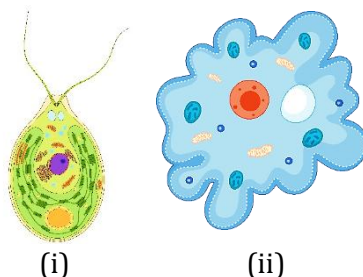
- (i) Bodies of the living beings are made up of cells. The cell is the basic unit of life.
- (ii) Activities of an organism are the sum total of activities of its cells.
- (iii) Every new cell arises from pre-existing cells.



Cell: The Building Block of Life DPP - 02

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

Directions: (Q.1 and Q.2) Observe the given image and answer the question.



- Identify the organism (i) -
 (1) amoeba (2) paramecium
 (3) chlamydomonas (4) euglena
- Which one of the following is unicellular?
 (1) (i) only (2) (ii) only
 (3) (i) and (ii) both (4) None of these
- Match the Column - I with Column - II.

Column - I		Column - II	
A	Paramecium	p	Multicellular
B	Plastid	q	Unicellular
C	Plant	r	Single membrane bound
D	Lysosome	s	Double membrane bound

Read the following statements and give your answer as true or false.

- Ribosome is non-membrane bound cell organelle.
- Mitochondria is a single membrane bound cell organelle.

Read the following statements and fill in the blanks.

- Chlamydomonas is an example of _____ organism.
- Cytoplasm includes _____ and _____.

Subjective Questions

- Draw a flow chart of structure of cell.
- Name two cell organelles which are covered by double membrane.
- Name two examples of unicellular organisms.

SOLUTIONS DPP-02

1. Option (3)

Organism (i) is chlamydomonas.

2. Option (3)

(i) and (ii) both are unicellular.

3. A -q, B- s, C- p, D- r

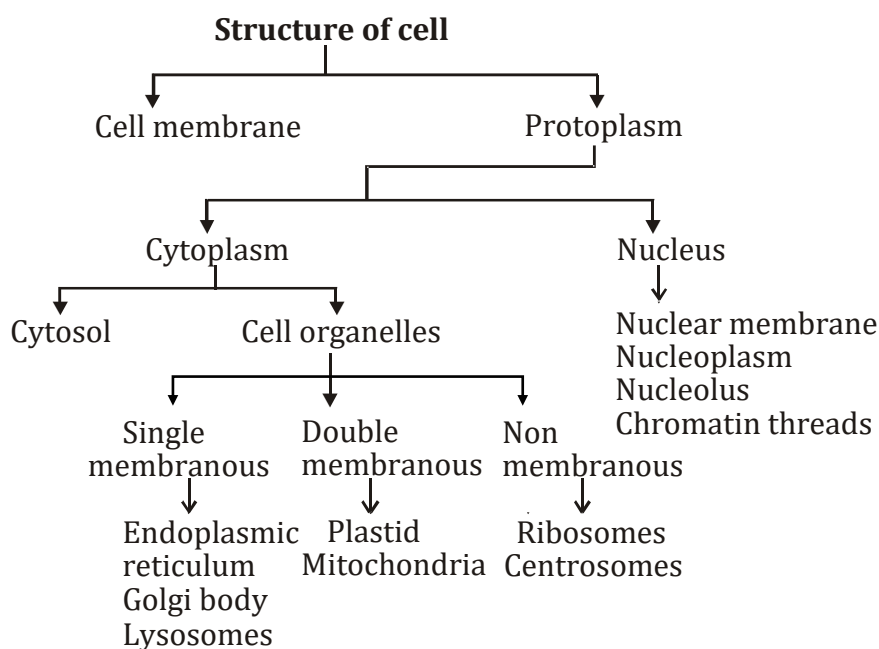
4. True

5. False

6. unicellular

7. cytosol, cell organelles

8.



9. (i) Mitochondria

(ii) Plastid

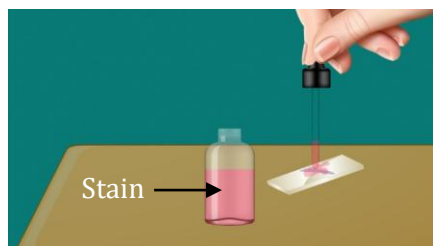
10. (i) Amoeba

(ii) Paramecium

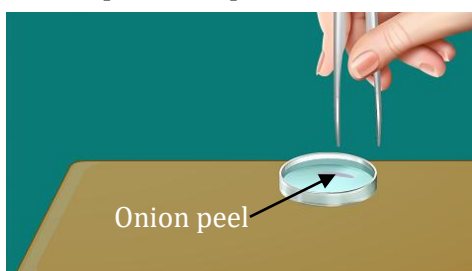
**Cell: The Building Block of Life DPP - 03**

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. The diagram is showing a step of onion peel experiment. Which stain is used here for staining the cells?



- (1) Methylene blue (2) Safranin
(3) Crystal violet (4) Ethanol
2. In the onion peel experiment, the peel is kept in water -



- (1) to keep it moist. (2) to stain it.
(3) to view it properly. (4) all of these

Read the following statements and give your answer as true or false.

3. Cheek cells having cell wall.
4. Onion peel cells have large vacuole.

Read the following statements and fill in the blanks.

5. _____ stain is used for staining the human cheek cell during the experiment.
6. Human cheek cells are flat and _____ in shape.

Subjective Questions

7. Why is glycerine used for mounting of the onion peel cells or human cheek cells?
8. Why is it essential to place the coverslip gently to avoid entry of air bubbles?
9. Why is staining done before mounting?
10. What are the precautions we should use during human cheek cell experiment?

SOLUTIONS DPP - 03

1. Option (2)

Safranin stain is used here for staining the cells.

2. Option (1)

The peel is kept in water to keep it moist.

3. False

4. True

5. Methylene blue

6. Polygonal

7. It prevents drying up the specimen.

8. It is essential to place the coverslip gently as some very soft specimens there is the possibility of damaging it. But the main purpose is to avoid the entry of air bubbles in the slide that hinders the clear vision.

9. Staining of specimen is done so as to get a better look it. Staining does more than just highlighting simply the outline of cells. Some stains can penetrate cell wall and highlight cell components and this helps also in visualising metabolic process. After all this is completed, than the specimen is mounted so that changes in the specimen can be seen for a period of time. Hence mounting is done after staining.

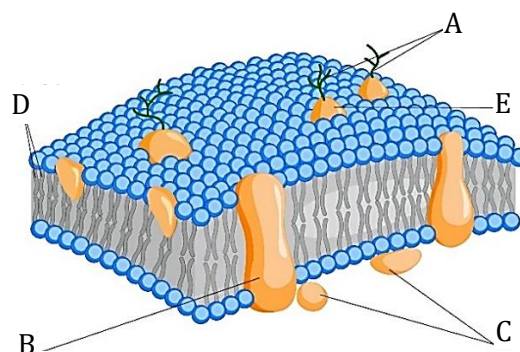
10. (i) Do not scrap the cheek too hard as it may injure the buccal lining.
(ii) Scrapped material should be spread uniformly on the slide.
(iii) Excess of stain should be drained off.
(iv) There should be no air-bubble under the coverslip.



Cell: The Building Block of Life DPP - 04

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

Directions: (Q.1 and Q.2) Observe the given image and answer the question.



- In the given structure the Hydrophobic portion of a lipid molecule is -
(1) A (2) B (3) C (4) D
- In the given diagram peripheral protein represents by -
(1) B (2) A (3) D (4) C
- Match the column - I with column - II.

Column - I		Column - II	
A	Freely movable	p	Selectively permeable
B	Plasma membrane	q	Oxygen
C	Fluid mosaic model	r	Singer and Nicolson
D	Amoeba	s	Endocytosis

Read the following statements and give your answer as true or false.

- Structure of plasma membrane was explained by Singer and Nicolson.
- Plasma membrane is selectively permeable.

Read the following statements and fill in the blanks.

- The plasma membrane is flexible and made up of organic molecules called _____ and _____.
- Plasma membrane allows materials to enter and leave the cell through the tiny holes called _____.

Subjective Questions

- What is active transport and passive transport?
- What are the functions of plasma membrane?
- Who gave the fluid mosaic model of plasma membrane?

SOLUTIONS DPP - 04

1. Option (4)

In the given structure the hydrophobic portion of a lipid molecule is D.

2. Option (4)

In the given diagram peripheral proteins represents by C.

3. A - q, B - p, C - r, D - s.

4. True

5. True

6. lipids, proteins

7. pores

8. Transport of substances across plasma membrane against the concentration gradient with expenditure of energy is called active transport and transport of substances across plasma membrane along the concentration gradient i.e. from higher concentration to lower concentration without expenditure of energy is called passive transport.

9 (i) It protects the internal components of the cell.

(ii) It provides shape to the cell.

(iii) It allows materials to enter and leave the cell through the tiny holes called pores.

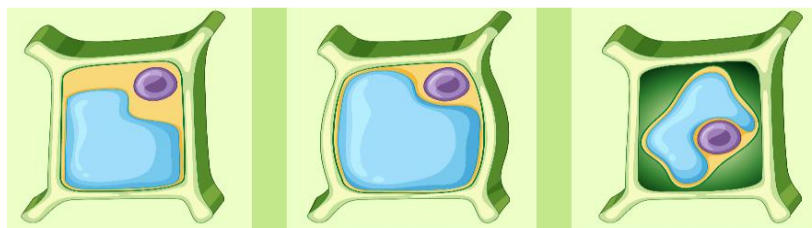
10. Singer and Nicolson



Cell: The Building Block of Life DPP - 05

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. The plant cell was placed in three different solutions named as A, B and C. Identify the nature of solution A, B and C.

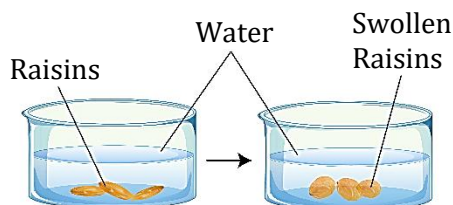


Normal cell placed
in Solution A

Normal cell placed
in Solution B

Normal cell placed
in solution C

- (1) A - Hypotonic, B - Isotonic, C - Hypertonic
 (2) A - Isotonic, B - Hypotonic, C - Hypertonic
 (3) A - Hypertonic, B - Isotonic, C - Hypotonic
 (4) A - Isotonic, B - Hypertonic, C - Hypotonic
2. What does following experimental setup depict?



- (1) Exosmosis
 (2) Endosmosis
 (3) Both (1) and (2)
 (4) Pinocytosis

Read the following statements and give your answer as true or false.

3. Absorption of water by plant root is the example of osmosis.
 4. Absorption of water by unicellular fresh water organism is the example of diffusion.

Read the following statements and fill in the blanks.

5. _____ is the osmotic entry of water into a cell.
 6. _____ is the osmotic withdrawal of water from a cell.

Subjective Questions

7. What is hypotonic solution?
 8. If the human blood cells are placed in hypotonic solution, it is observed that they burst. Give reasons.
 9. When a perfume bottle opens in a room, it spread uniformly in the room. Which process responsible for this?
 10. What is diffusion?

SOLUTIONS DPP-05

1. **Option (2)**

Solution A - Isotonic solution,

Solution B - Hypotonic solution

Solution C - Hypertonic solution

2. **Option (2)**

The given experimental setup depicts endosmosis.

3. True

4. False

5. Endosmosis

6. Exosmosis

7. The external solution is dilute as compared to cell contents. It has more water content while the water content is lower inside the cell. The cell membrane allows passage of water in both directions. Due to difference in concentration of water molecules, there is net flow of water molecules into the cells. The phenomenon is called endosmosis.

8. If the human blood cells are placed in hypotonic solution, it is observed that they burst, because human blood cells lack cell wall.

9. Diffusion

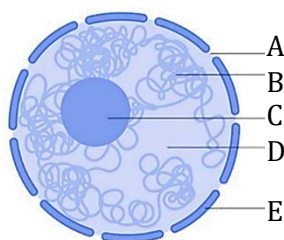
10. The process of movement of substance (solid, liquid & gas) from the region of its higher concentration to the region of its lower concentration to spread uniformly in the given space is called diffusion.



Cell: The Building Block of Life DPP - 06

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

- Nucleus is discovered by -
 (1) Robert Hooke (2) Leeuwenhoek
 (3) Robert Brown (4) Fontana
- From the diagram given below, identify the parts labelled as B and C.



- (1) Nucleoplasm, Centrosome (2) Cytoplasm, Lysosome
 (3) Cytoplasm, Nucleus (4) Chromatin material, Nucleolus
- Match the Column - I with Column - II.

Column I		Column II	
A	Plant cell wall	p	Chitin
B	Bacterial cell wall	q	Cellulose
C	Fungi cell wall	r	Peptidoglycan

Read the following statements and give your answer as true or false.

- Cell wall is a protective and supportive coat.
- Cell wall provides definite shape to the cell.

Read the following statements and fill in the blanks.

- Cell wall is not present in _____.
- A cell can exist without _____ like animal cell but cannot exist without cell membrane.

Subjective Questions

- What is the structure of cell wall in plants?
- Write a short note on cellulose.
- Draw the diagram of plasmolysis.

SOLUTIONS DPP - 06

1. Option (3)

Nucleus is discovered by Robert Brown.

2. Option (4)

The part labelled as B is chromatin material and C is nucleolus.

3. A -q, B -r, C - p

4. True

5. True

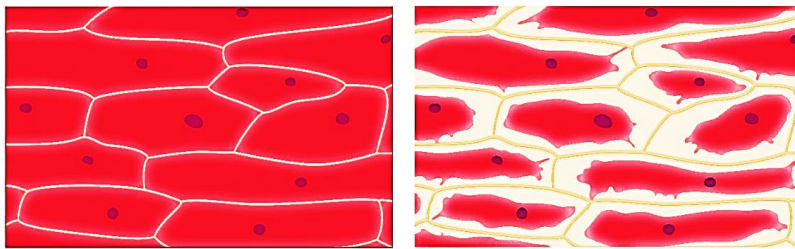
6. animal cell.

7. cell wall

8. Cell wall of plant cells is formed by a fibrous polysaccharide called cellulose, while it is formed by peptidoglycan in bacteria and blue-green algae. It is formed by chitin in most of fungi.

9. Cellulose is a complex fibrous carbohydrate which cannot be digested by human beings and several other animals but can be digested by ruminant cattle due to presence of bacteria in their gut which secrete cellulase enzyme.

10.



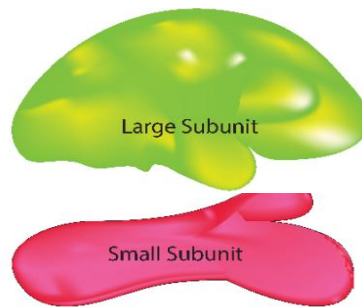
Plasmolysis



Cell: The Building Block of Life DPP - 07

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. Identify the picture given below.



- (1) Ribosome (2) Lysosome
(3) Endoplasmic reticulum (4) Golgi body
2. What is the function of ribosome?
- (1) Lipid synthesis (2) Protein synthesis
(3) Carbohydrate synthesis (4) Nucleic acid synthesis

Read the following statements and give your answer as true or false.

3. Ribosome is also called engine of the cell.
4. Each ribosome is made up with two equal sub-units which join together at the time of protein synthesis.

Read the following statements and fill in the blanks.

5. The prokaryotic ribosome is made up of _____ and _____ sub-units.
6. The eukaryotic ribosome is made up of _____ and _____ sub-units.

Subjective Questions

7. 'Organelle within an organelle' term is used for?
8. Cytoplasm consists of how many components/parts? Name them.
9. Define cytoplasm.
10. What is svedberg's unit?

SOLUTIONS DPP - 07

1. Option (1)

It is ribosome.

2. Option (2)

Function of ribosome is protein synthesis.

3. True

4. False

5. 50S, 30S

6. 60S, 40S

7. Ribosome is known as 'organelles within an organelle'.

8. Cytoplasm is made up of two components known as cytosol & cell organelles.

9. The fluid content/protoplasmic mass of the cell inner to plasma membrane and excluding nucleus is called cytoplasm.

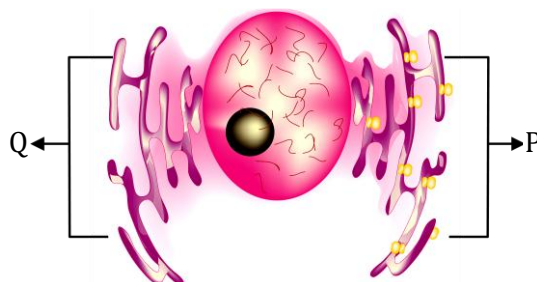
10. 'S' (Svedberg's unit) stands for the sedimentation coefficient; it is indirectly measure the density and size.



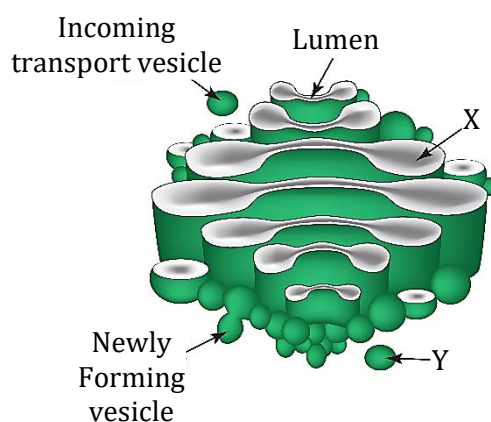
Cell: The Building Block of Life DPP - 08

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. Diagram of endoplasmic reticulum is given below. Identify the labelled structure P and Q.



- (1) P - RER, Q - SER
(2) P - SER, Q - RER
(3) P - RER, Q - RER
(4) P - SER, Q - SER
2. Figure of Golgi body is given below. Identify the labelled structure X and Y.



- (1) X - Cisternae, Y - Tubules.
(2) X - Tubules, Y - Vesicles.
(3) X - Cisternae, Y - Vesicles.
(4) X - Vesicles, Y - Tubules.
3. Match the Column - I with Column - II.

Column - I		Column - II	
A	RER	p	Detoxification
B	SER	q	Golgi body of plants
C	Golgi body	r	Protein synthesis
D	Dictyosomes	s	Camillo Golgi

Read the following statements and give your answer as true or false.

4. Golgi complex was discovered by Camillo Golgi in the nerve cells of dog.
5. Golgi bodies are pleomorphic structures.

Read the following statements and fill in the blanks.

6. _____ is the site of formation of glycolipids and glycoprotein.
7. _____ plays an important role in detoxification of drugs and poisons.

Subjective Questions

8. Which organelle is associated with the surface of RER?
9. What are the functions of endoplasmic reticulum?
10. What do you understand by the term “membrane biogenesis”?

SOLUTIONS DPP - 08

1. Option (1)

P - Rough endoplasmic reticulum

Q. - Smooth endoplasmic reticulum

2. Option (3)

X - Cisternae

Y - Vesicles

3. A - r, B - p, C - s, D - q

4. False

5. True

6. Golgi complex

7. Smooth endoplasmic reticulum

8. Ribosomes

9. (i) ER serve as channels for the transport of materials between various regions of cytoplasm or between cytoplasm and nucleus.

(ii) It also functions as a cytoplasmic framework providing a surface for some of the biochemical activities of the cell.

(iii) SER plays an important role in detoxifying many poisons and drugs and is also a site of lipid synthesis.

(iv) RER is engaged in synthesizing proteins and enzymes.

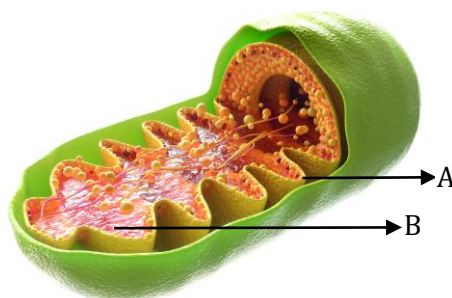
10. The proteins and lipids synthesized by RER and SER respectively help in building the cell membrane which is known as membrane biogenesis.



Cell: The Building Block of Life DPP - 09

Q.1. is multiple choice question. It has four options (1), (2), (3) and (4) out of which only one option is correct.

1. Identify the labelled structure in the diagram given below.



- (1) A - Matrix, B - Oxysomes
 (2) A - Matrix, B - Oxysomes
 (3) A - Cristae, B - Matrix
 (4) A - Matrix, B - Cristae
2. Match the column - I with column - II.

Column - I		Column - II	
A	Lysosome	p	Power house of cell
B	Mitochondria	q	Suicidal bags
C	ATP	r	Tennis racket shaped
D	F ₁ particles	s	Energy currency

Read the following statements and give your answer as true or false.

3. Mitochondria are the double membrane bound cell organelle.
 4. Lysosomes are called semi-autonomous cell organelle.

Read the following statements and fill in the blanks.

5. _____ of mitochondria increase the inner surface area to hold a variety of enzymes.
 6. Mitochondria has _____ type of ribosome.

Subjective Questions

7. Draw the diagram of mitochondria.
 8. Why are lysosomes known as "suicidal bags"?
 9. Write a short note on origin of mitochondria.
 10. What are the functions of mitochondria?

SOLUTIONS DPP - 09

1. Option (3)

A - Cristae

B - Matrix

2. A - q, B - p, C - s, D - r

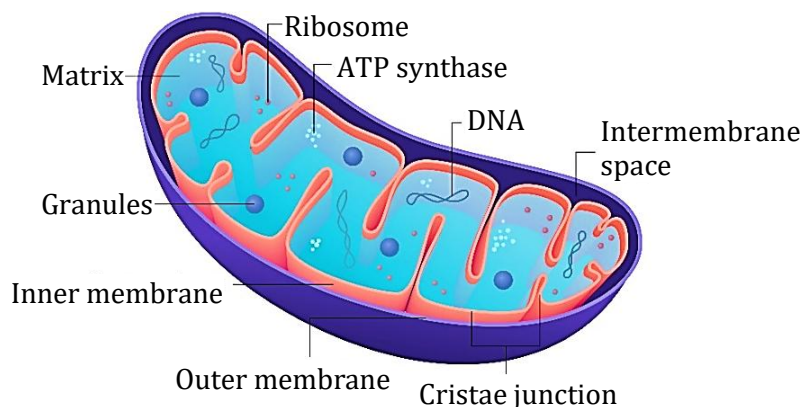
3. True

4. False

5. Cristae

6. 70 s

7.



8. At the time of cellular damage, the lysosomes burst to release its enzyme and they digest its own cell. Therefore, lysosomes are also known as suicidal bags of a cell.

9. Mitochondria have a small and circular chromosome and 70S ribosome of their own and make some of their own proteins, showing that they were once aerobic bacteria.

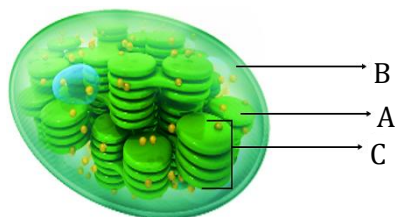
10. Mitochondria are the main sites of cellular respiration. They bring about complete oxidation of food stuffs or respiratory substrates into carbon dioxide and water.

They are commonly known as 'power houses of the cell' because they contain enzymes necessary for the complete oxidation of food and for release of high amount of energy in the form of ATP (Adenosine triphosphate) molecules. The body uses energy stored in ATP for synthesis of new chemical compounds and for mechanical work. ATP is also known as energy currency of the cell.

**Cell: The Building Block of Life DPP - 10**

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. Identify the labelled structure A, B and C in the diagram given below.



- (1) A - Stroma, B - Thylakoid, C - Grana
(2) A - Stroma, B - Grana, C - Thylakoid
(3) A - Grana, B - Stroma, C - Thylakoid
(4) A - Thylakoid, B - Stroma, C - Grana
2. Chlorophyll pigment is present inside -
- (1) Stroma (2) Thylakoid
(3) Inner membrane (4) None of these

Read the following statements and give your answer as true or false.

3. Chromoplasts are red, yellow and orange in colour and are found in petals of flower and in fruits.
4. Chloroplasts are usually disc -shaped and surrounded by a single membrane.

Read the following statements and fill in the blanks.

5. Chloroplasts are the sites of _____.
6. Plastid has _____ type of ribosomes.

Subjective Questions

7. Draw the diagram of plastid.
8. Write the name of different types of plastids.
9. There would be no animal life if chloroplasts did not exist. Justify.
10. Why are plastids termed as semi - autonomous?

SOLUTIONS DPP - 10

1. Option (4)

A - Thylakoid

B - Stroma

C - Grana

2. Option (2)

Chlorophyll pigment is present inside thylakoid.

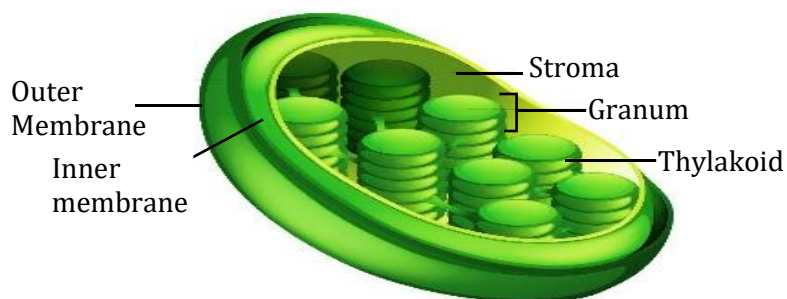
3. True

4. False

5. photosynthesis

6. 70s

7.



8. (i) Chromoplasts

(ii) Leucoplasts

(iii) Chloroplast

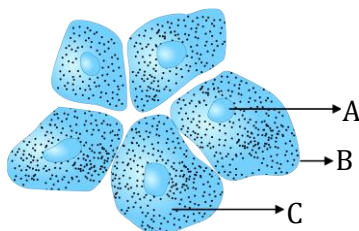
9. If chloroplasts did not exist, there would be no plant life because if chloroplast did not exist then there will be no oxygen for animals and there will be no food for animals. So, survival of animals cannot be possible.

10. Plastid is semi-autonomous cell organelle as they are having their own DNA and ribosomes which are useful in protein synthesis.

**Cell: The Building Block of Life DPP - 11**

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

Directions: (Q.1 and Q.2) Observe the given image and answer the question.



1. Identify the structures A, B and C
 - (1) A - Nucleus, B - Cytoplasm, C - Plasma membrane
 - (2) A - Plasma membrane, B - Cytoplasm, C - Nucleus
 - (3) A - Nucleus, B - Plasma membrane, C - Cytoplasm
 - (4) A - Cytoplasm, B - Plasma membrane, C - Nucleus
2. Which of the following structure is present in both animal and plant cell?
 - (1) A
 - (2) A and B
 - (3) A, B and C
 - (4) Only A and C

Read the following statements and give your answer as true or false.

3. Centrioles are present only in higher plants.
4. Cell wall present in plant cell.

Read the following statements and fill in the blanks.

5. _____ act as dump house for excretory products in plants cells.
6. Chloroplast present in _____ cell.

Subjective Questions

7. Write four differences between animal and plant cell.
8. What is tonoplast?
9. What is cell sap?
10. What are the functions of vacuoles?

SOLUTIONS DPP - 11

1. Option (3)

A - Nucleus

B - Plasma Membrane

C - Cytoplasm

2. Option (3)

A, B and C, represent the nucleus, cell membrane, cytoplasm and these all are present in both animal and plant cell.

3. False

Centrioles are not present plants.

4. True

Cell wall present in plant cell.

5. Vacuoles

6. Plant

7. Comparison between Animal cell and Plant cell

Animal Cell		Plant Cell
Cell wall	Absent	Present
Shape of the cell	Round (irregular shape)	Rectangular (fixed shape)
Chloroplast	Animal cells don't have chloroplast	Plant cells have chloroplasts because they make their own food
Nucleus	Present in center	Present in periphery

8. The covering membrane of the vacuole is called tonoplast.

9. The fluid content of the vacuole is called cell sap. Vacuoles store amino acid, sugars, various organic acids and some proteins.

10. (i) In plant cell, vacuoles are full of cell sap and provide turgidity and rigidity to the cell.

(ii) Vacuoles store amino acid, sugars, various organic acids and some proteins.

(iii) They act as dump house for excretory products in plant cells.

(iv) In single celled organisms like Amoeba, the food vacuole contains the food items that the amoeba has consumed.

**Cell: The Building Block of Life DPP - 12**

Q.1. and Q.2 are multiple choice questions. Each has four options (1), (2), (3) and (4) out of which only one option is correct.

1. Meiosis -
 - (1) required for repair for body.
 - (2) required to replace old, dead and injured cells.
 - (3) produces four new cells.
 - (4) all of these
2. I am a cell organelle exclusively found in animal cells and lower plant forms. I am associated with cell division. Who am I?
 - (1) Ribosomes
 - (2) Lysosomes
 - (3) Centriole
 - (4) Vacuole

Read the following statements and give your answer as true or false.

3. The process of cell division by which most of the cells divide for growth is called mitosis.
4. Meiosis helps in growth and repair of tissues in organisms.
5. When a cell divides by meiosis it produces four new cells.

Subjective questions

6. What is cell division?
7. What are the two main types of cell division?
8. Write a short note on mitosis.
9. Write a short note on meiosis.
10. Which type of cell division helps in growth and repair of tissues?

SOLUTIONS DPP - 12**1. Option (3)**

In meiosis cells divide to produce four new cells while in mitosis it is two.

2. Option (3)

Centriole is the cell organelle exclusively found in animal cells and lower plant forms. Centriole is associated with cell division by forming spindle fibre.

3. True

The process of cell division by which most of the cells divide for growth is called mitosis.

4. False

Mitosis helps in growth and repair of tissues in organisms.

5. True

When a cell divides by meiosis it produces four new cells.

6. The process by which new cells are made is called cell division.

- 7.** (i) Mitosis
(ii) Meiosis

8. The process of cell division by which most of the cells divide for growth is called mitosis. In this process, each cell called mother cell divides to form two identical daughter cells. The daughter cells have the same number of chromosomes as mother cell. It helps in growth and repair of tissues in organisms.

9. Specific cells of reproductive organs or tissues in animals and plants divide to form gametes, which after fertilisation give rise to offspring. They divide by a different process called meiosis which involves two consecutive divisions. When a cell divides by meiosis it produces four new cells instead of just two. The new cells only have half the number of chromosomes than that of the mother cells.

10. Mitosis.